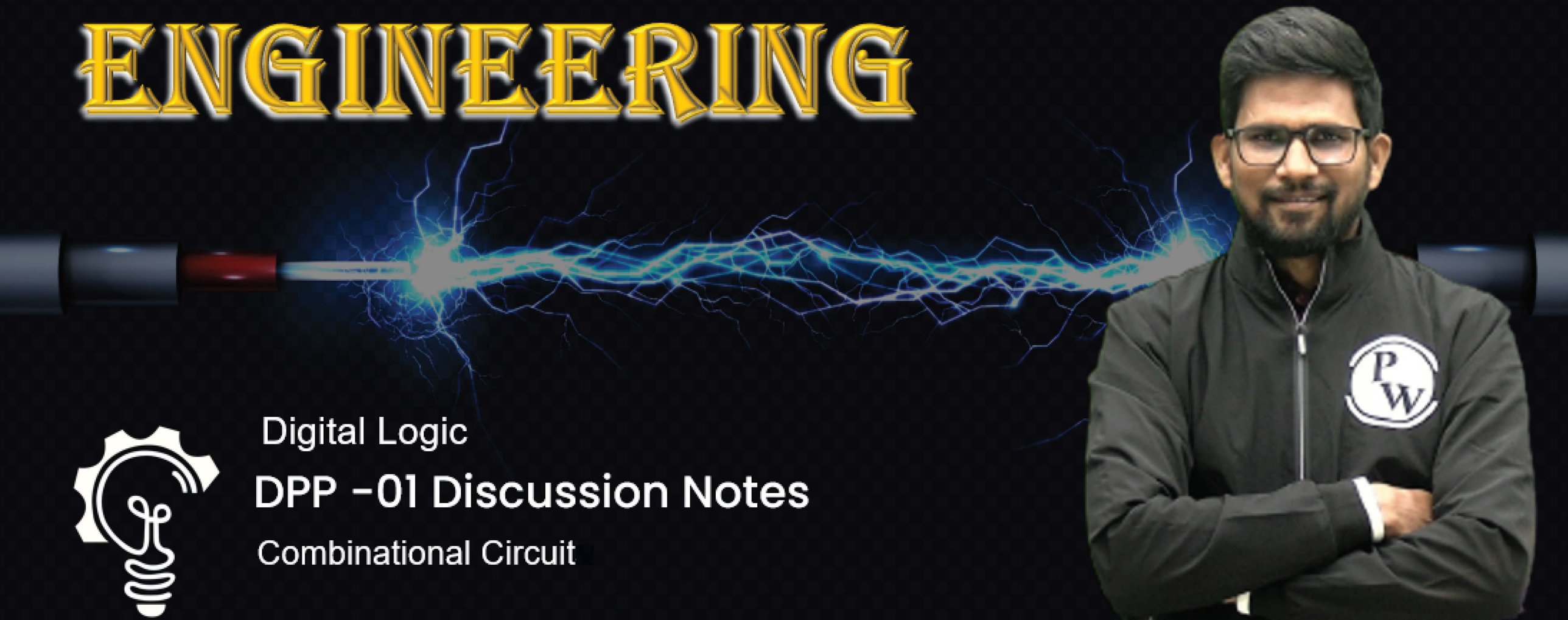
CS & IT ENGINEERING



Digital Logic

DPP -01 Discussion Notes

Combinational Circuit

By- Chandan Gupta Sir

Question (MCQ) EASY

Which of the following is true?

A All gates follow commutative as well as associative law. ✗

B NOR & XNOR does not follow associative law. ✗

C NAND & XOR does not follow associative law ✗

D ✓ All gates follow commutative law but some gates doe not follow associative law.

Question (MCQ) MODERATE

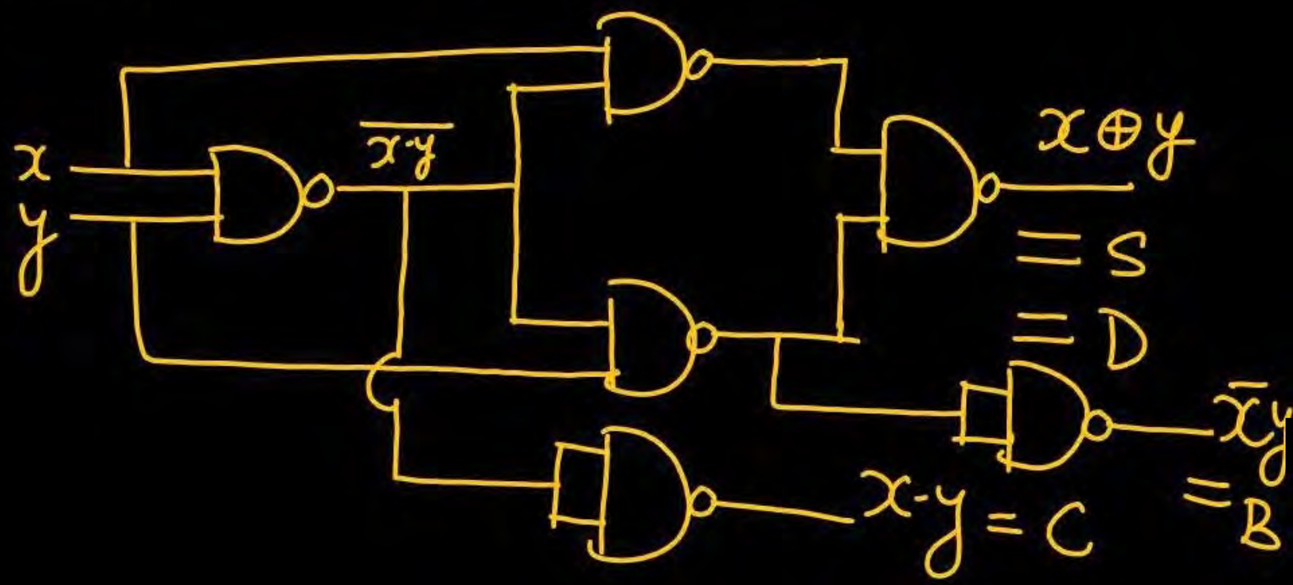
We have to implement half adder as well as half subtractor. Minimum number of 2-i/P NAND gate require is 6.

H·A· ⟶ $S = x \oplus y$

$C = x \cdot y$

H·S ⟶ $D = x \oplus y$

$B = \bar{x} y$



Question (MCQ) EASY

A digital circuit is as given below:

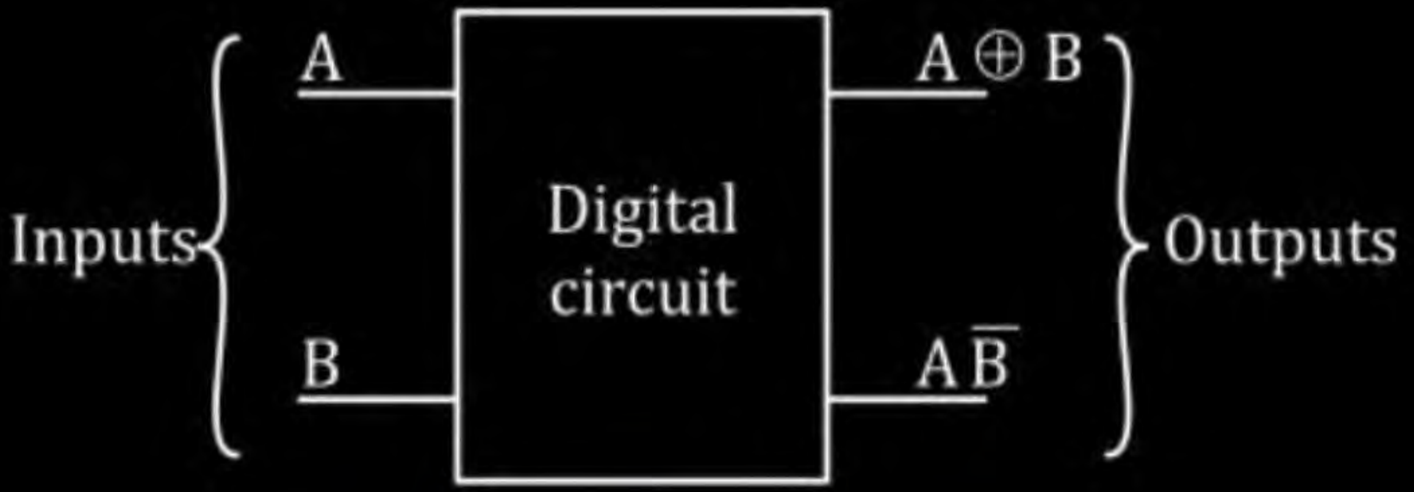
The circuit implements

A $A + B$

B $A - B$

C $B - A$

D None of these

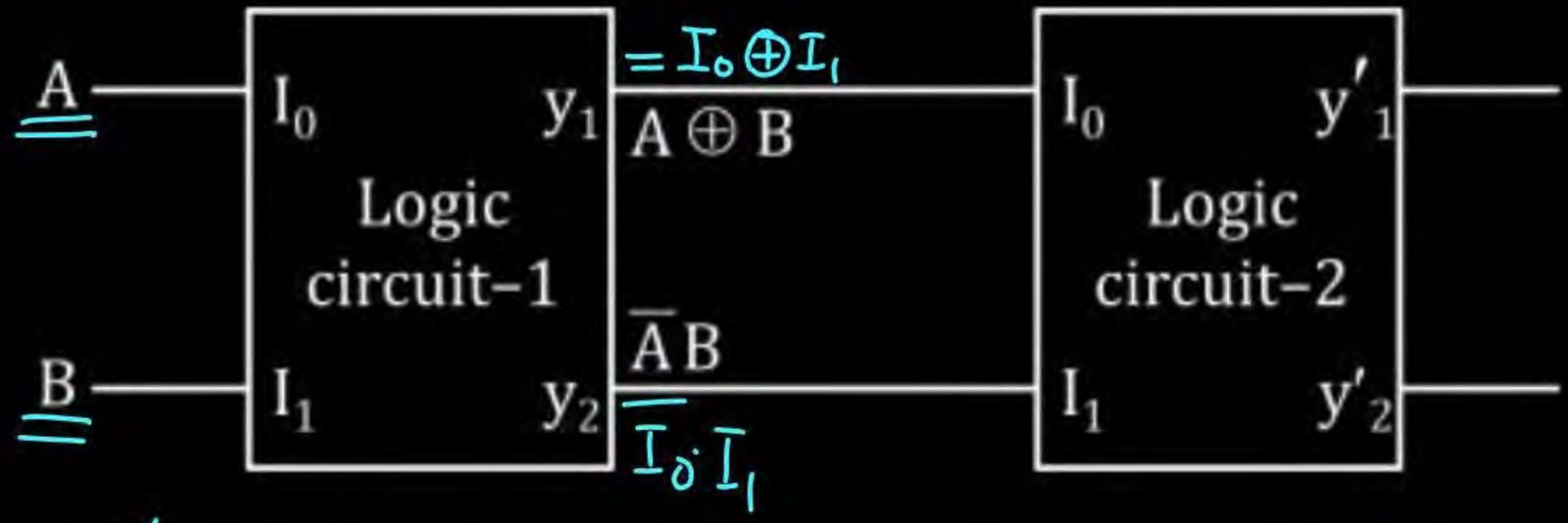


$A - B \Rightarrow \text{Borrow} = \bar{A} \cdot B$

$B - A \Rightarrow \text{Borrow} = \bar{B} \cdot A = A\bar{B}$

Question (MCQ) MODERATE

A logical circuit is as given below:



Function of logic circuit – 1 & logic circuit – 2 is same. Then output y'_1 will be

A $A \oplus B$

B $A\bar{B}$

C $\bar{A}B$

D $A - B$

$y_1' = (A \oplus B) \oplus \bar{A}B$

$= A \oplus B \oplus \bar{A}B = A \oplus [\bar{B} \cdot \bar{A}B + B \cdot \overline{\bar{A}B}]$

$= A \oplus [B \cdot (A + \bar{B})] = A \oplus [AB]$

$y_1' = A \oplus AB = \bar{A}AB + A \cdot \overline{AB} = A \cdot [\bar{A} + \bar{B}]$

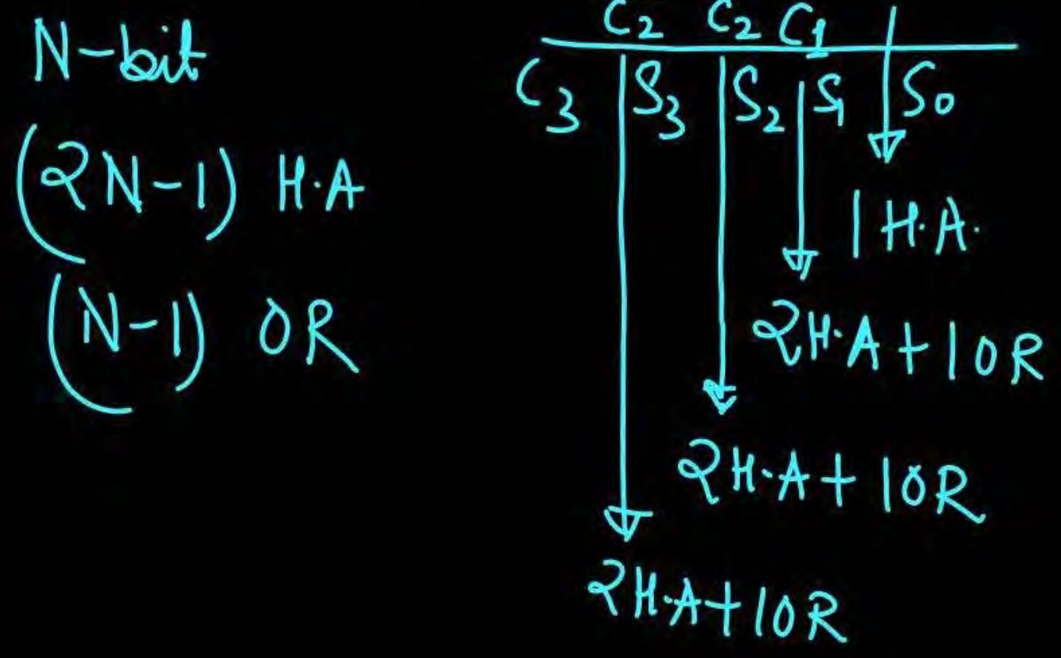
$= 0 + A\bar{B} = A\bar{B}$

Question (MCQ) EASY

We have two numbers A and B and both are 4-bit numbers. To perform complete addition of these two numbers, total number of H.A.s + OR gate require is 10.

$A = a_3 a_2 a_1 a_0$,

$B = b_3 b_2 b_1 b_0$



N-bit

(2N-1) H.A

(N-1) OR

H.A's $n = 7$ H.A

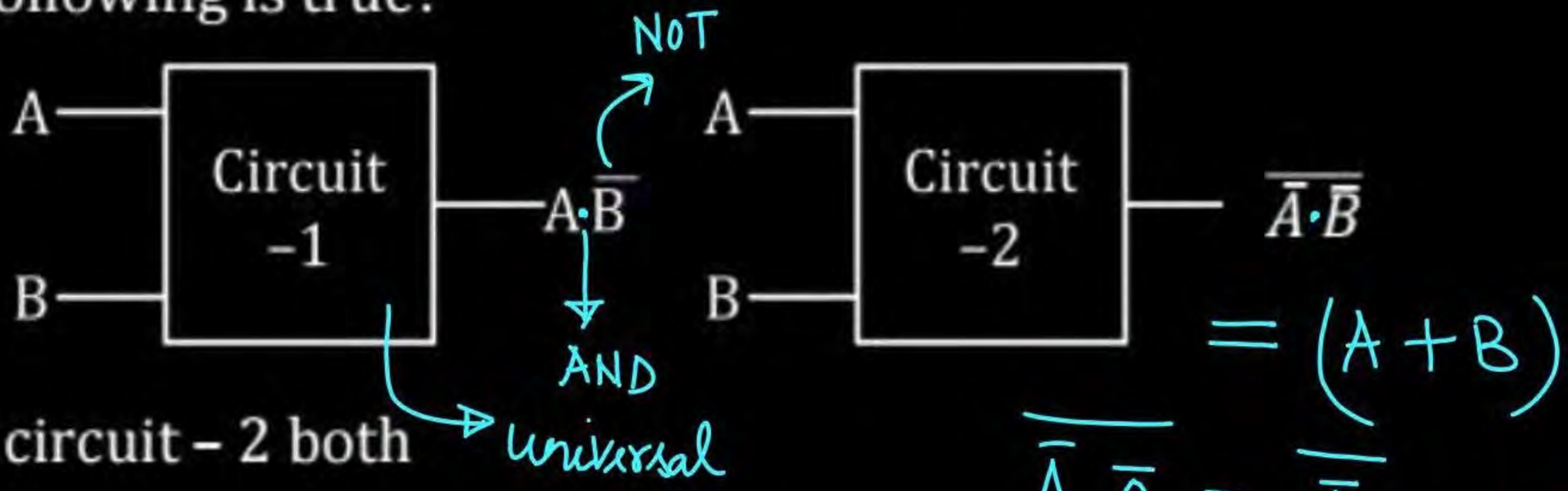
OR gate $m = 3$ OR

$n + m = 7 + 3 = 10$

Question (MCQ)

BRAIN TEASER

Which of the following circuits can be used as universal circuit :
Then which of the following is true?



$= (A + B)$

$\overline{\bar{A} \cdot \bar{0}} = \overline{\bar{A}} = A$

$\overline{\bar{A} \cdot \bar{1}} = \overline{0} = 1$

$\overline{\bar{A} \cdot \bar{A}} = \overline{\bar{A}} = A$

A circuit – 1 and circuit – 2 both

B circuit – 1 only

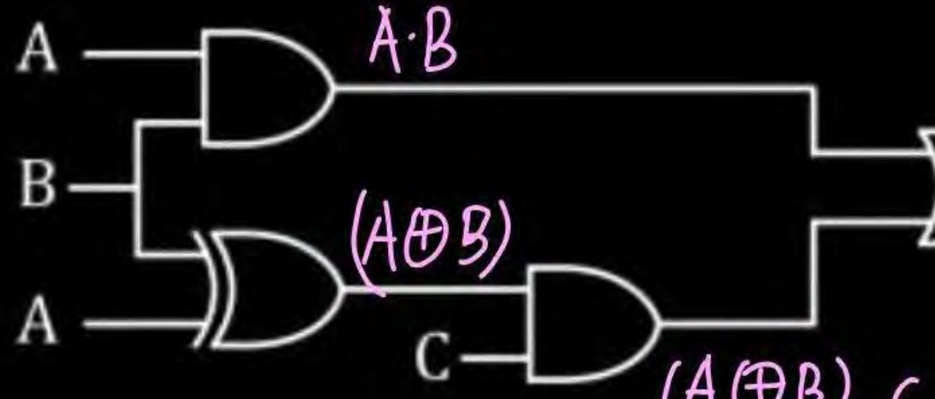
C circuit – 2 only

D Neither circuit – 1 nor circuit – 2

Question (MSQ)

BRAIN TEASER

A logical circuit is implemented as :
Output y is



$y = AB + (A \oplus B)C$

Carry o/p = $AB + BC + CA$

(A, B)

A. '1' when majority of the i/P$_s$ A, B, C are at '1'

B. '0' when minority of the i/P$_s$ A, B, C are at '1'

C. '1' when minority of the i/P$_s$ A, B, C are at '1'

D. '0' when majority of the i/P$_s$ A, B, C are at '1'

A	B	C
1	0	0
0	1	0
0	0	1
0	0	0

Question (MCQ) BRAIN TEASER

x, y, z are i/Ps of a full adder then, sum output of full adder will be '1' if

$$\text{Sum} = x \oplus y \oplus z = \Sigma(1,2,4,7)$$

A $\bar{x}\bar{y} + \bar{y}\bar{z} + \bar{z}\bar{x} = 1$ ✗

B $xyz = 1$ ✗

C $(x + y)(y + z)(z + x) = 0$ ✗

D ✓ None of these

Sum = 1 ← { only one '1' & two zeros / all 3 i/p are at 1.

$$\overline{(x+y)\cdot(y+z)\cdot(z+x)} = \overline{0}$$

$$(\overline{x+y}) + (\overline{y+z}) + \overline{z+x} = 1$$

$$(\bar{x}\cdot\bar{y}) + (\bar{y}\cdot\bar{z}) + \bar{z}\cdot\bar{x} = 1$$

Question (MSQ) (B, C) EASY



$= A + B$

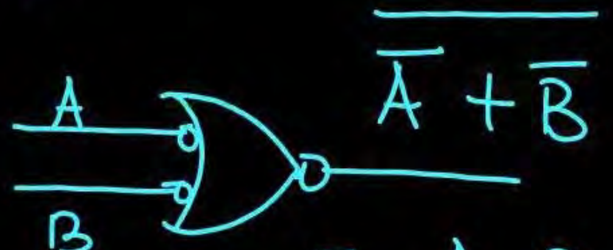
Which of the following is/are true?

A NOR gate is equivalent to NAND gate with bubbled i/P$_s$. ✗

B OR gate is equivalent to NAND gate with bubbled i/P$_s$. ✓

C AND gate is equivalent to NOR gate with bubbled i/P$_s$. ✓

D ✗ NAND gate is equivalent to NOR gate with bubbled i/Ps.



$= A \cdot B$

Question (MCQ) MODERATE

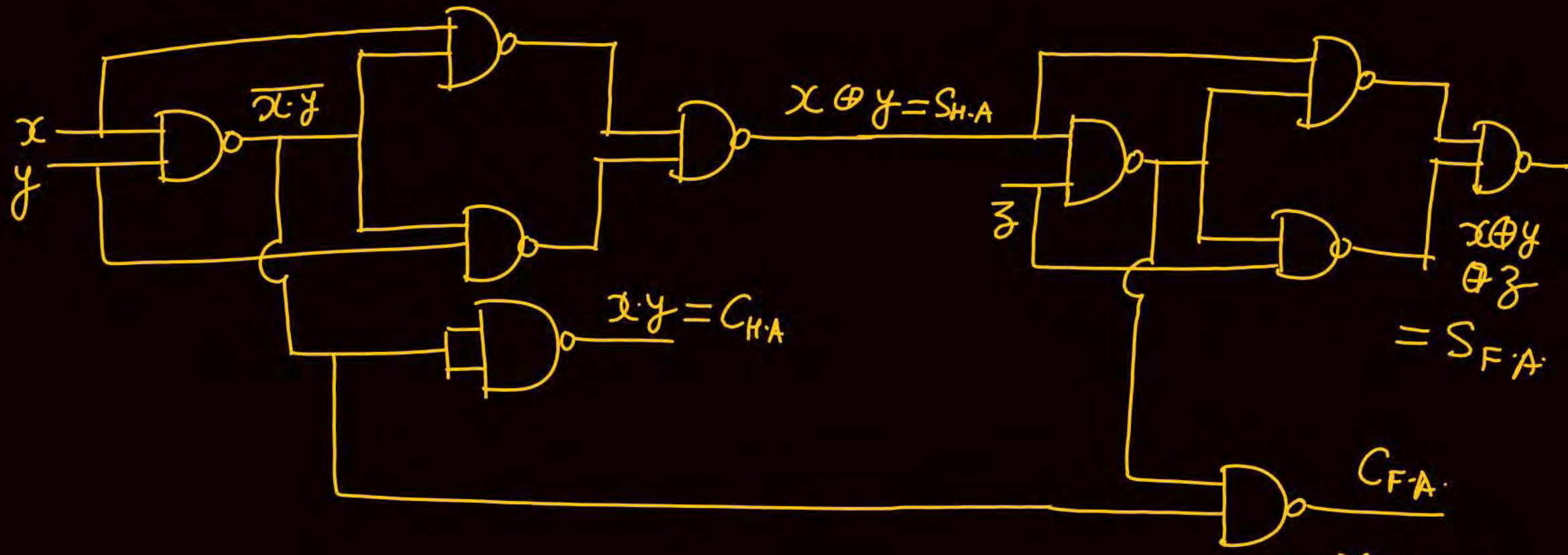
We have to implement H.A circuit & full adder circuit together. i/P lines for H.A. is x, y and i/P lines for full adder is x, y, z then minimum number of 2 – i/P NAND gate require is 10.

$H.A \rightarrow S_{H.A} = x \oplus y$

$C_{H.A} = xy$

$F.A \rightarrow S_{F.A} = x \oplus y \oplus z$

$C_{F.A} = xy + (x \oplus y) \cdot z = xy + yz + zx.$


x
y
$\overline{x \cdot y}$
$x \oplus y = S_{H\cdot A}$
$x \cdot y = C_{H\cdot A}$
z
$x \oplus y \oplus z = S_{F\cdot A}$
$C_{F\cdot A}$
$= xy + z(x \oplus y)$

Question NAT

We have a 4-bit adder in which each full adder has sum delay of t_s = 4 nsec and carry delay of 2 nsec. In one minute number of additions completed by this full adder is 6 X 10^9

$t_s = 4$ nsec, $t_c = 2$ nsec

Worst case delay $t_w = (n-1)\,t_c + \max[t_c, t_s] = 3t_c + \max(t_c, t_s)$

$= 3 \times 2 + \max(2\text{ns}, 4\text{ns}) = 10$ nsec

∴ 1 addition is performed in 10 nsec

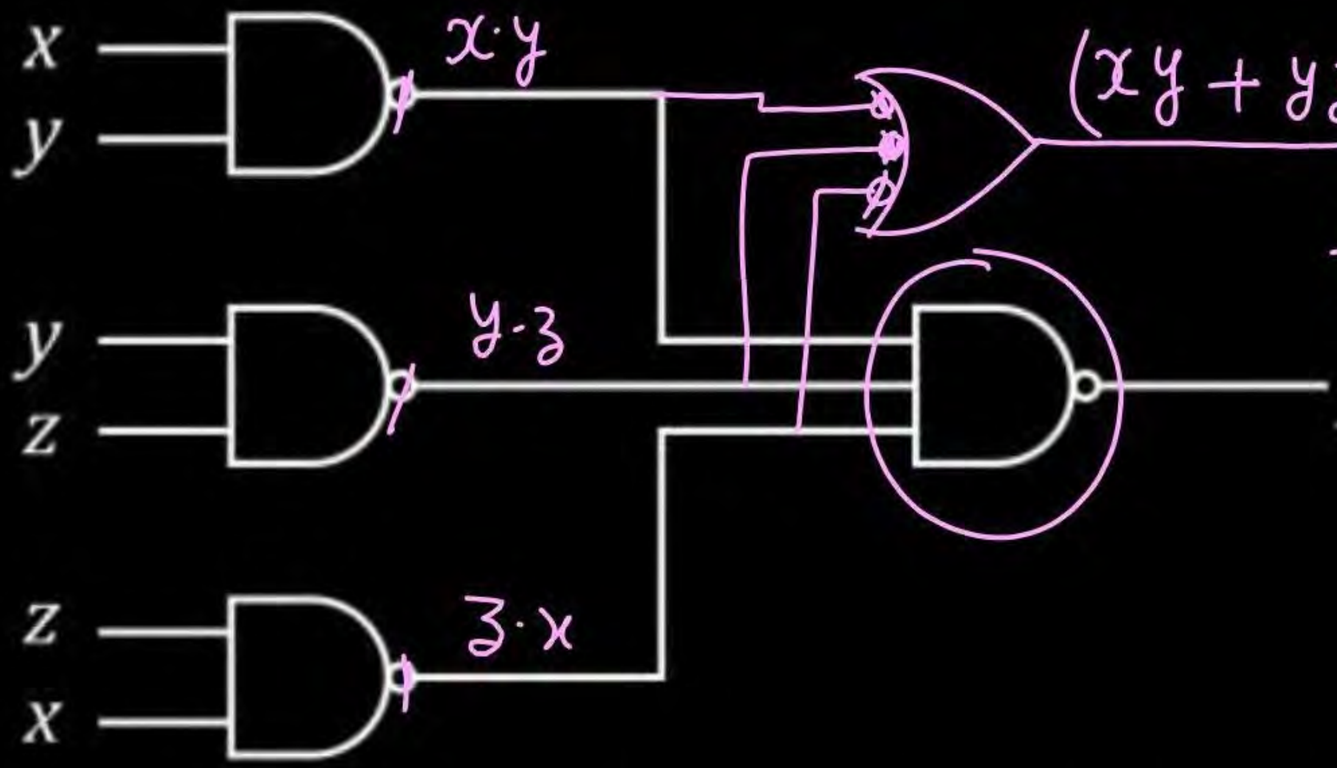
∴ no. of addition performed in one $= \frac{1\text{ sec}}{10\text{ nsec}} = \frac{10^9}{10} = 10^8$ additions

∴ no. of addition in 1 mitute (60sec) $= 60 \times 10^8 = 6 \times 10^9$ additions

Question MSQ (A, C)

★★☆

A logical circuit is as given below:



$(xy + yz + zx) = \text{Carry o/p} = f_c$

$f_c^D = (x+y)\cdot(y+z)\cdot(z+x)$

$f(x, y, z) = xy + yz + zx$

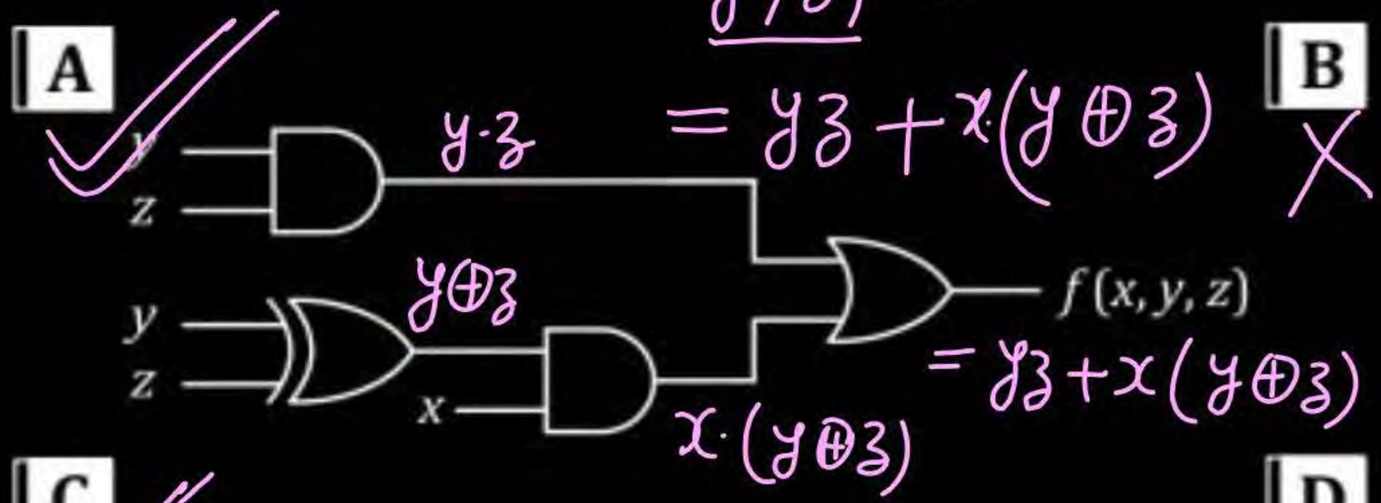
$= f_c$

Which of the following circuit is/are equivalent to the above circuit:

i/p y, z, previous bit generated carry x

$f = yz + zx + xy$

A



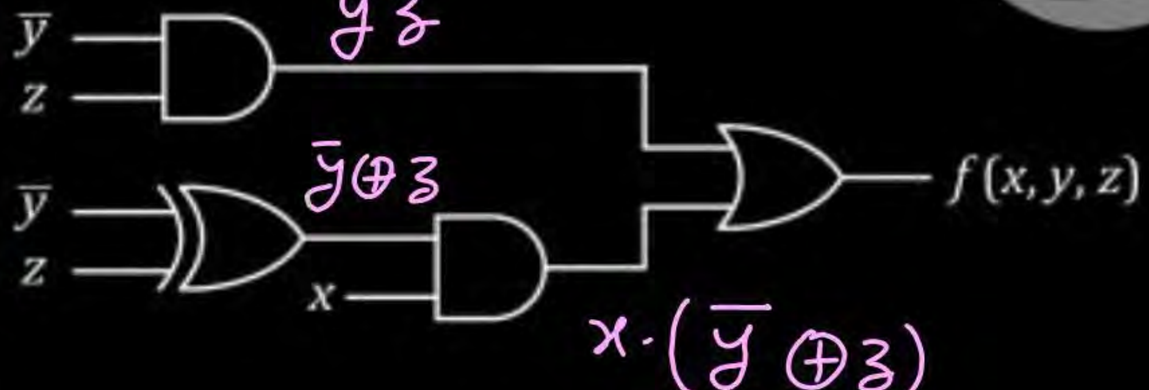
$y \cdot z$

$y \oplus z$

$x \cdot (y \oplus z)$

$= yz + x(y \oplus z)$

B ✗



$\bar{y}z$

$\bar{y} \oplus z$

$x \cdot (\bar{y} \oplus z)$

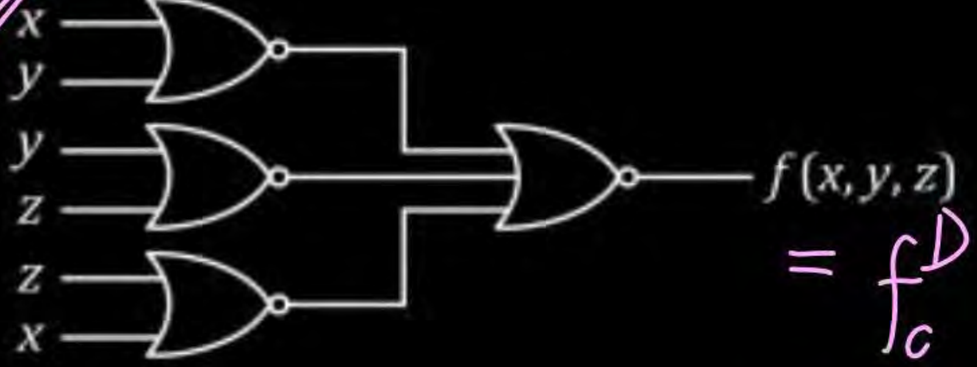
$= \bar{y}z + x \cdot (y \oplus z)$

$(y - z)$ i/p borrow bit x

$\bar{y}z + x(y \odot z)$

$y \quad z \quad x$

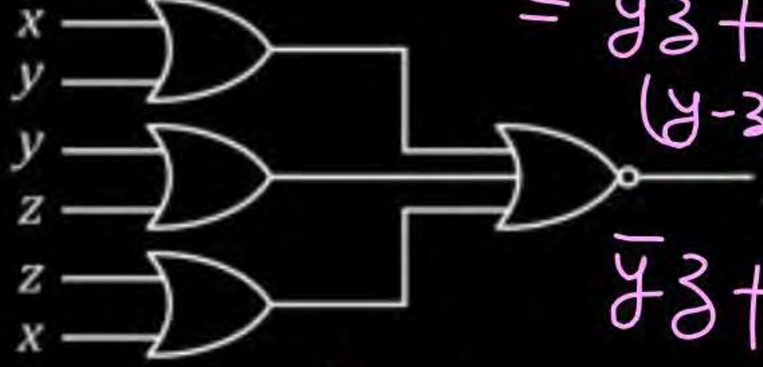
C



$= f_c^D$

$= f_c$

D ✗



Question MCQ

A K-map is as given below :

In minimized solution of above K-map, there are terms $\bar{x}\bar{y}$, $\bar{x}\bar{z}$ and one term is missing. The missing term is

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1	1		1
x		1	1	

$\bar{x}\bar{y}$ $\bar{x}\bar{z}$ xz

A $\bar{y}z$

B xz

C $\bar{x}z$

D $\bar{y}\bar{z}$

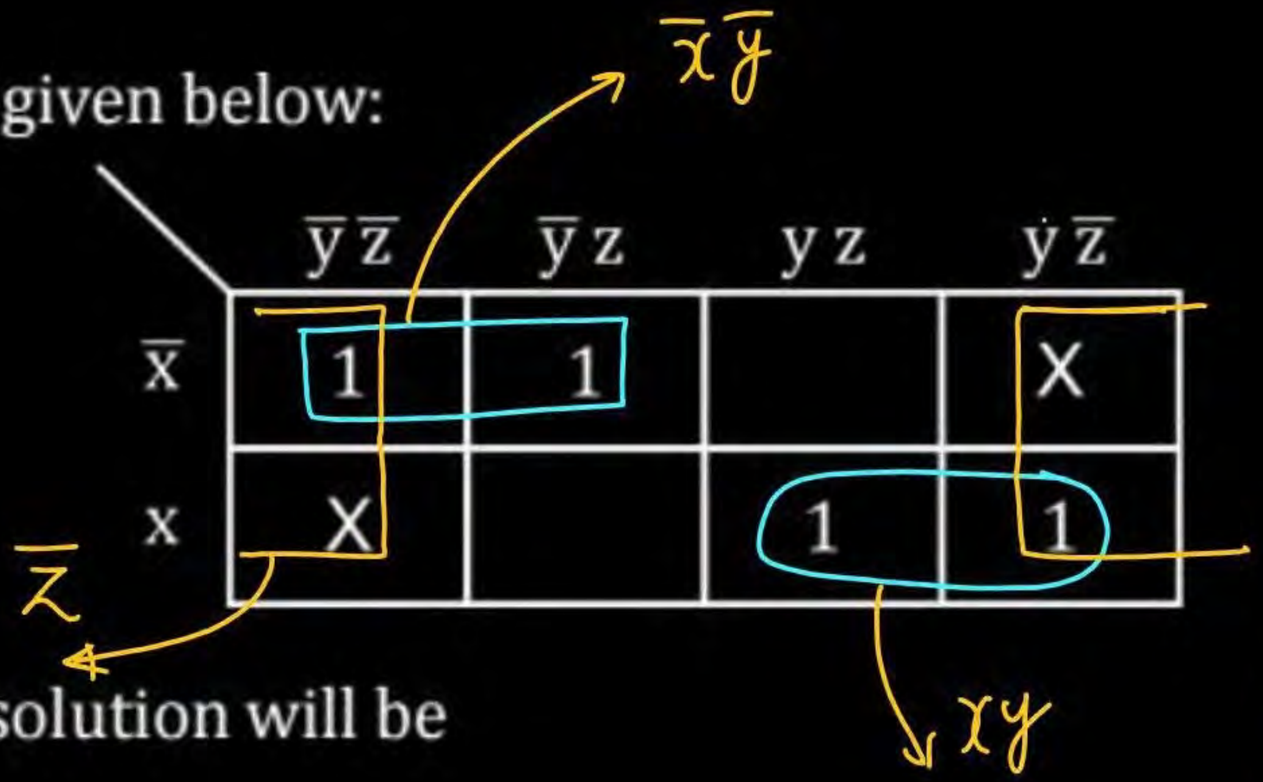
	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1	1		1
x		1	1	

$\bar{x}\bar{z}, xz, \bar{y}z$
or
$\bar{x}\bar{z}, xz, \bar{x}\bar{y}$

Question

MCQ

A K-Map is as given below:



$= \bar{x}\bar{y} + xy$

$= x \odot y$

It minimized solution will be

A $x \odot y$

B $x \oplus y$

C $\bar{z} + \bar{x}\bar{y}$

D $\bar{z} + xy + \bar{x}\bar{y}$

Question

MCQ

$F(x, y, z) = \overline{x\bar{y} + \bar{x}y} + x\bar{y}z + x\bar{y}\bar{z}$ $= \overline{f_1} + x\bar{y}z + x\bar{y}\bar{z}$

Then the minimized solution of F(x,y,z) will be

$f_1 = x\bar{y} + \bar{x}y$

$\overline{F} = x + \bar{y}$

A	$x + \bar{y}$
B	$\bar{x}y$
C	$xy\bar{z}$
D	$x + \bar{y} + z$

$\overline{f_1}$	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1	1	0	0
x	0	0	1	1

F	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1	1		
x	1	1	1	1

$\bar{y}$ x

Question

NAT



In a 4-bit parallel adder, sum delay and carry delay of each full adder is t_s = 1 nsec and t_c = 5 nsec respectively. Number of additions performed per sec by adder is N_1. If delays are interchanged then no. of additions performed per sec is N_2 then N_2/N_1 2.5.

$n = 4$ bits $\longrightarrow$ $t_s = 1$ ns, $t_c = 5$ nsec $\longrightarrow$ N_1 additions in one second

Worst delay $t_w = (n-1)t_c + \max(t_c, t_s) = 3t_c + \max(5\text{ns}, 1\text{ns})$

$t_w = 20$ nsec

$N_1 = \frac{1 \text{ sec}}{t_w} = \frac{1 \text{ sec}}{20 \text{ nsec}} = \frac{1}{20 \times 10^{-9}} = \frac{10^9}{20}$ additions

$\therefore \frac{N_2}{N_1} = \frac{10^9/8}{10^9/20} = 2.5$

$t_s' = 5$ nsec, $t_c' = 1$ nsec $\rightarrow$ $t_w' = 3t_c' + \max(t_c', t_s')$

$= 3 \times 1 + \max(1\text{nse}, 5\text{ns})$

$= 8$ nse

$\therefore N_2 = \frac{1 \text{ sec}}{t_w'} = \frac{1 \text{ sec}}{8 \text{ nse}} = \frac{10^9}{8}$ addition

Question

MCQ

A K-map is as given below :

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	1	
$\bar{A}B$	1	1	1	X
AB			1	
$A\bar{B}$		X	X	

$= \bar{A}\bar{C} + CD$

Then minimized solution of K-map is :

A $\bar{A}\bar{C}+\bar{A}D+CD$

B $\bar{A}\bar{C}+CD$

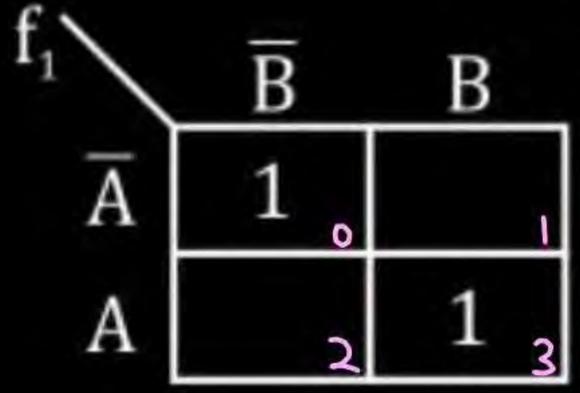
C $\bar{A}B+CD+\bar{A}\bar{C}$

D None of these

Question

MCQ

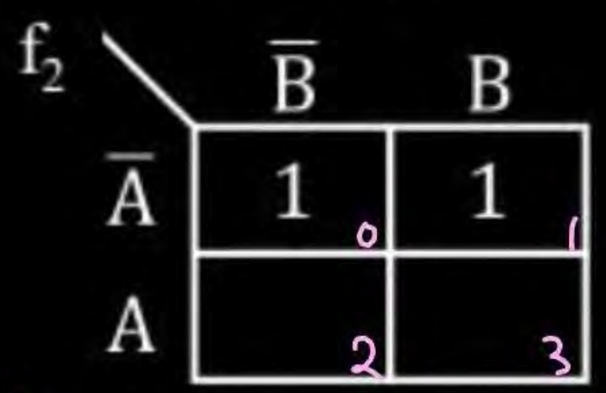
K-map of two logical functions f_1 (A, B) and f_2 (A, B) is as given below :



$f_1(A,B) = \Sigma(0,3)$

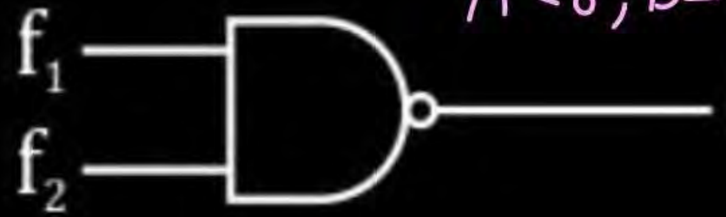
A=0, B=0

A=0, B=1



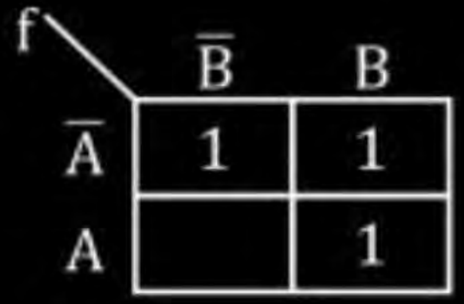
$f_2(A,B) = \Sigma(0,1)$

Then K-map of f (A, B) will be

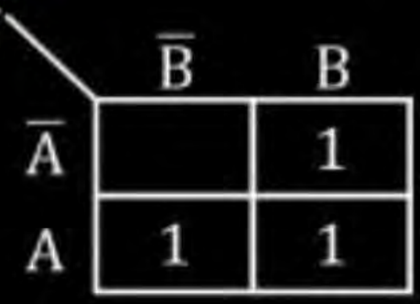


$f(A,B) = \Sigma(1,2,3)$

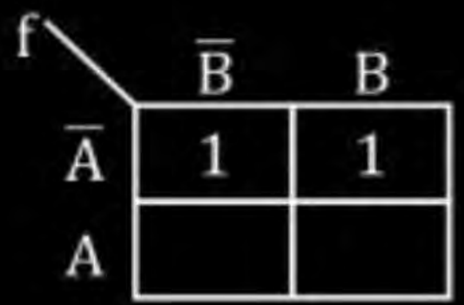
A


f
$\overline{B}$
B
$\overline{A}$
A
1
1
1

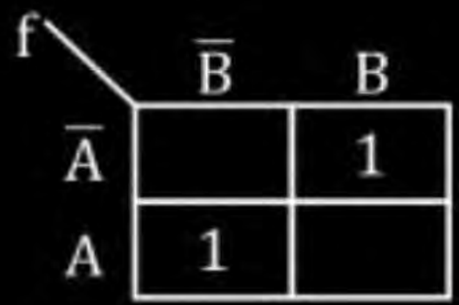
B


f
$\overline{B}$
B
$\overline{A}$
A
1
1
1

C


f
$\overline{B}$
B
$\overline{A}$
A
1
1

D


f
$\overline{B}$
B
$\overline{A}$
A
1
1

Question

NAT

★★☆

f (A, B, C, D) is as given below :

$$f(A,B,C,D)=\overline{\bar{B}D+AC+\bar{A}B}+ABCD+\bar{A}BC+ABD = \overline{f_1}+ABCD+\bar{A}BC+ABD$$

Then number of entries in K-map of f (A, B, C, D) that are filled with '1'

8

$f_1 = \bar{B}D + AC + \bar{A}B$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1			1
$\bar{A}B$			1	1
AB	1	1	1	
$A\bar{B}$	1			

f_1

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	0	0	1
$\bar{A}B$	0	0	0	0
AB	1	1	0	0
$A\bar{B}$	1	0	0	0

Question

MCQ

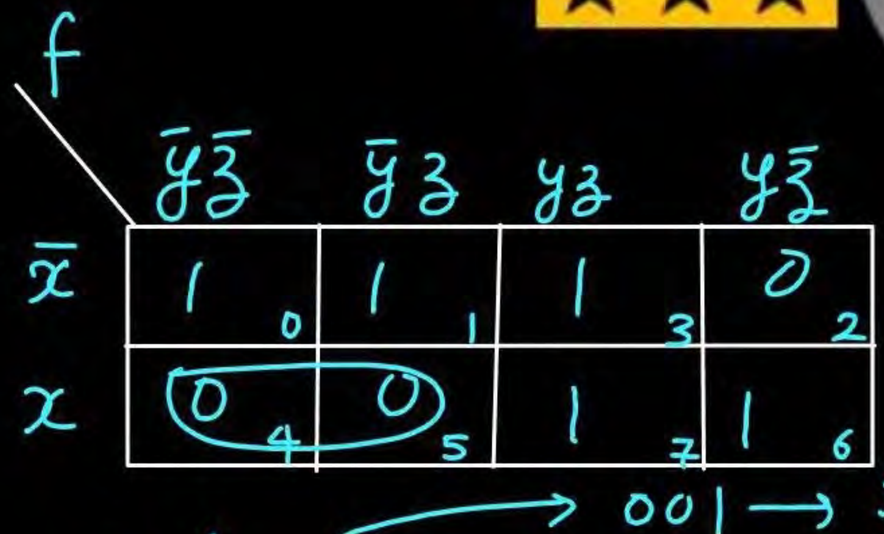
f (x, y, z) is given as : $f(x,y,z)=\bar{x}\bar{y}+\bar{x}z+xy$,
The equivalent f (x, y, z) can also be

A $(\bar{x}+y+\bar{z})(x+\bar{y}+\bar{z})(x+\bar{y}+z)$

B $(x+\bar{y}+z)(\bar{x}+y+z)(\bar{x}+y+\bar{z})$

C $(\bar{x}+\bar{y})(\bar{x}+z)(x+y)$

D None of these



$f(x,y,z)=\Sigma(0,1,3,6,7)=\Pi(2,4,5)$

010, 100, 101

$=(x+\bar{y}+z)\cdot(\bar{x}+y+z)(\bar{x}+y+\bar{z})$

Question

MCQ

Which of the following is true ?

A K-map has more than one solution. ✗

B K-map has always unique solution. ✗

C K-map can have more than one solution. ✓

D None of these

Question NAT

$A > B = 1 + 2 + 3 = 6$

Two no's A & B are two bit no's as

$A = a_1a_0$ and $B = b_1b_0$

Then no. of combination in which $A > 2B$ ____4____.

A → 2 bit → 0, 1, 2, 3

B → 2 bit → 0, 1, 2, 3

2B → 0, 2, 4, 6

When A = 0 → no combination
A > 2B

When A = 1 → one combination
A > 2B [when 2B = 0]

When A = 2 → one combination
A > 2B [2B = 0]

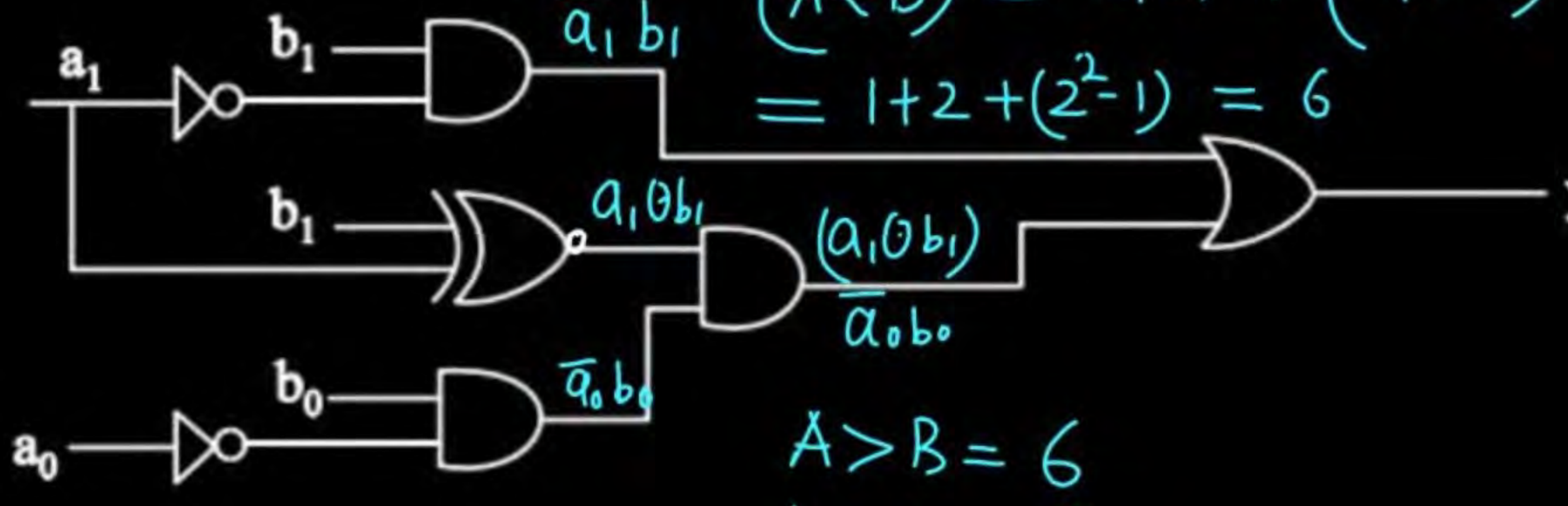
When A = 3 → two combinations
A > 2B [2B = 0, 2B = 2]

Total = 1 + 1 + 2 = 4

Question NAT

★★★

A digital ckt is as given below:



$A = a_1 a_0$

$B = b_1 b_0$

$(A<B) = \bar{a}_1 b_1 + (a_1 \odot b_1)\, \bar{a}_0 b_0$

$= 1 + 2 + (2^2 - 1) = 6$

$A > B = 6$

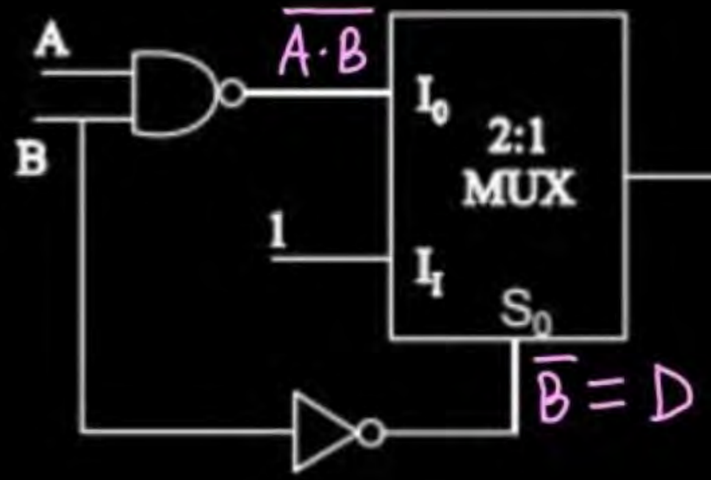
$A = B = 2^n = 4$

i/p lines of the ckt are a_1, b_1, a_0, b_0 and o/p line is y.
Then no. of combinations of i/p lines in which o/p y will be '1' is 6.

Question MCQ

★★☆

A MUX circuit is as given below:



o/p y is.

$y = \overline{D} \cdot \overline{A \cdot B} + D \cdot 1$

$= B(\overline{A} + \overline{B}) + \overline{B}$

$= \overline{A} \cdot B + 0 + \overline{B}$

$= \overline{A}B + \overline{B} = \overline{B} + (\overline{A} \cdot B)$

$= (\overline{B} + \overline{A}) \cdot (\overline{B} + B) = (\overline{A} + \overline{B})$

$= \overline{A \cdot B}$

A $\overline{A} + \overline{B}$

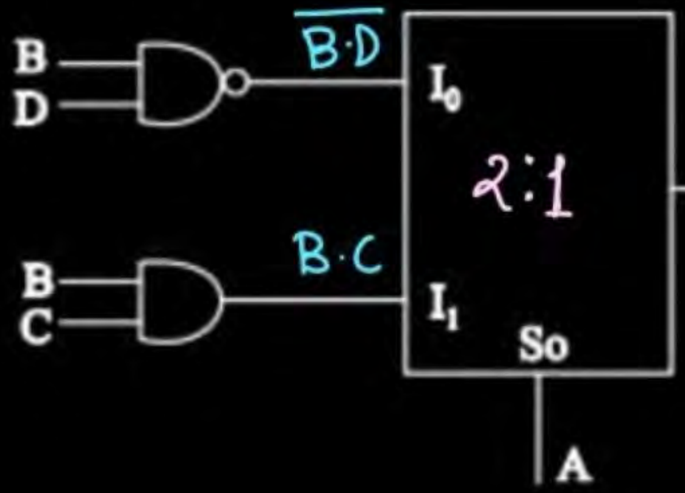
B $\overline{A + B}$

C $A \oplus B$

D $A\overline{B}$

Question MCQ

A MUX ckt is as given below:



$$f = \bar{A}\,\overline{B \cdot D} + A \cdot BC$$
$$= \bar{A}(\bar{B} + \bar{D}) + ABC$$
$$= \bar{A}\bar{B} + \bar{A}\bar{D} + ABC$$

f(A, B, C, D) is:

A $\Sigma(0, 1, 2, 3, 4, 5, 6, 14, 15)$

B $\Sigma(0, 1, 2, 3, 4, 6, 13, 15)$

C $\Sigma(0, 1, 2, 3, 4, 6, 14, 15)$

D $\Sigma(1, 2, 3, 4, 5, 6, 7, 13, 14, 15)$

$\bar{A}\bar{B}$

0 0 0 0 → 0
0 0 0 1 → 1
0 0 1 0 → 2
0 0 1 1 → 3

$\bar{A}$ $\bar{D}$

0 0 0 0 → 0
0 0 1 0 → 2
0 1 0 0 → 4
0 1 1 0 → 6

$\Sigma(0,1,2,3,4,6,14,15)$

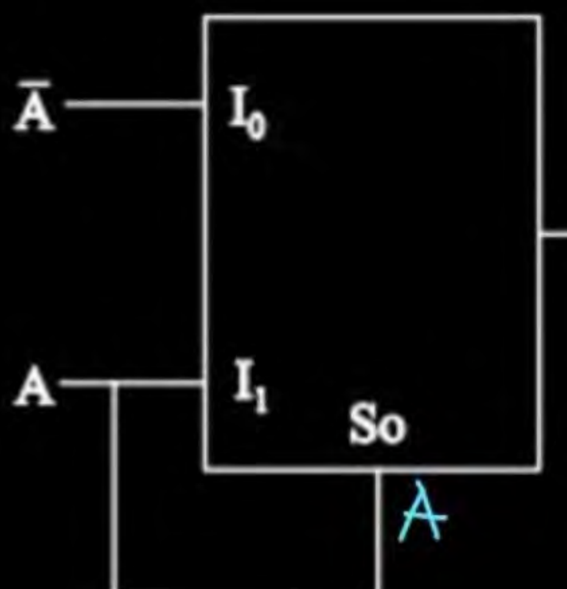
A B C

1 1 1 0 → 14
1 1 1 1 → 15

Question MCQ

★☆☆

A MUX ckt is as given below
o/p y is



$y = A \odot S_0$
$= A \odot A$
$= 1$

$= \bar{S}_0 \cdot \bar{A} + S_0 \cdot A$
$= \bar{A} \cdot \bar{A} + AA$
$= A + \bar{A} = 1$

A	$\bar{A}$	B	A
C	0	D	1

Question MCQ

★★☆

$y = 1 \oplus B = \overline{B}$

A 4:1 MUX is given as:

D	$\overline{x}$	C	A
$\overline{D}$	x	$\overline{C}$	$\overline{A}$
$\overline{D}$	x	$\overline{C}$	$\overline{A}$
D	$\overline{x}$	C	A



$y = C \oplus S_1 \oplus S_0 = A \oplus \bar{A} \oplus B$

$= \overline{S_1}\,\overline{S_0}\cdot C + \overline{S_1} S_0 \cdot \overline{C} + S_1 \overline{S_0}\,\overline{C} + S_1 S_0 \cdot C$

$f(S_1, S_0, C) = \sum(1,2,4,7)$

$= S_1 \oplus S_0 \oplus C$

$\overline{x} = D$

$x = \overline{D}$

o/p y is

$= D \oplus S_1 \oplus S_0$

$= \overline{x} \oplus S_1 \oplus S_0$

$= \overline{x \oplus S_1 \oplus S_0}$

A $\overline{B}$

B B

C $A \oplus B$

D $A \odot B$

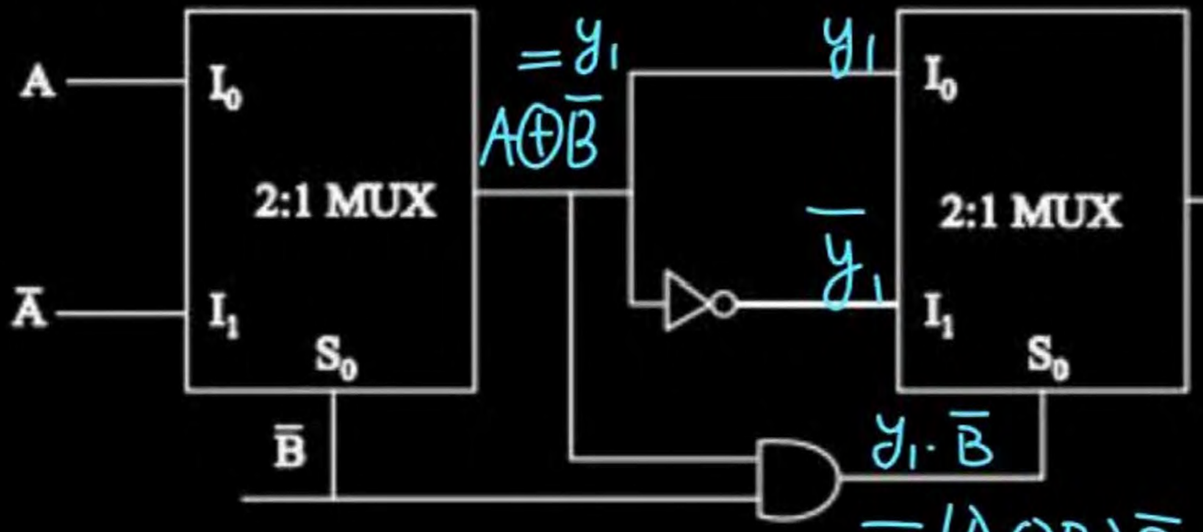
Question MCQ

★★★

$A \oplus B \oplus AB = (A+B)$

$A \oplus B \oplus (A+B) = A \cdot B$

A combinational ckt is designed using MUX as shown below :-



$y = y_1 \oplus S_0 = A \oplus \bar{B} \oplus \overline{(A+B)}$

$= A \oplus B \odot \overline{(A+B)}$

$= A \oplus B \oplus (A+B)$

$= A \cdot B$

$= (A \odot B)\bar{B} = (AB + \bar{A}\bar{B})\bar{B} = \bar{A}\bar{B} = \overline{A+B}$

o/p y is

A $\overline{A+B}$

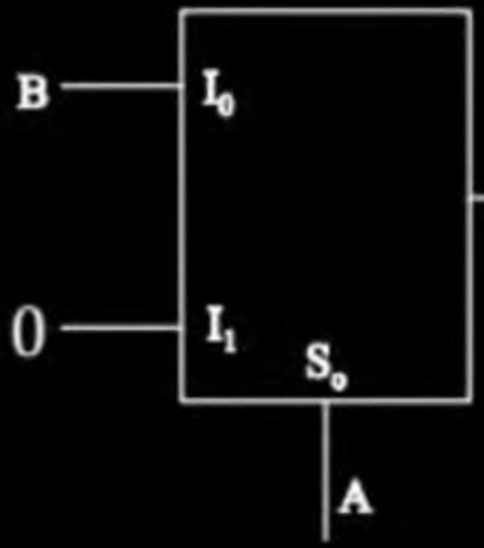
B A.B

C $A \oplus B$

D $A \odot B$

Question MCQ

A MUX circuit is implemented as shown:



$y = \bar{A}B + A \cdot 0$

$y = \bar{A} \cdot B$

o/p y represents

$(A < B) = \bar{A}B$

$(A > B) = A\bar{B}$

$(A = B) = A \odot B$

A Comparator o/p when B < A

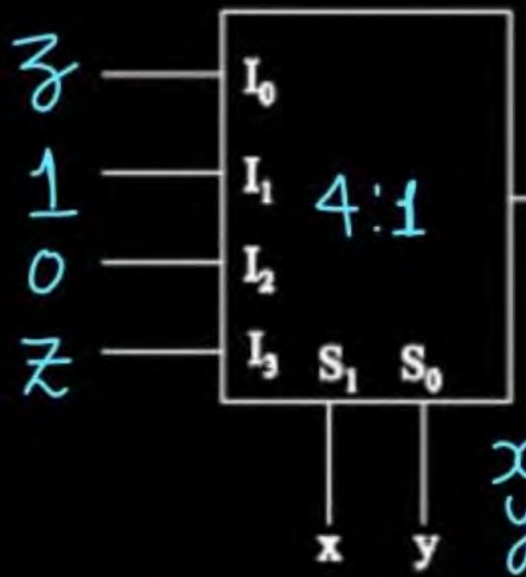
B Comparator o/p when B = A

C Comparator o/p when B > A

D Comparator o/p when B < 2A

Question MCQ

In a full subtractor corresponding i/p bits of two no.s are x & y [x – y] and input borrow is z and output borrow is B. Output borrow is implemented as shown :



$B = \bar{x}y + \bar{x}z + yz$

$B(x=0, y=0) = z + 0 = z$

$y = \bar{x}y + \bar{x}z + yz$

$B(x=0, y=1)$

$= 1 + \overline{\quad}$

$= 1$

$x=0$, $y=0$] $x=0$, $y=1$

Then values of I_0, I_1, I_2 & I_3 are :

A ✓ $I_0 = z, I_1 = 1, I_2 = 0, I_3 = z$

B ✗ $I_0 = z, I_1 = 0, I_2 = 1, I_3 = \bar{z}$

C ✗ $I_0 = 1, I_1 = z, I_2 = 0, I_3 = z$

D ✗ $I_0 = \bar{z}, I_1 = 1, I_2 = 0, I_3 = z$

Question NAT





To implement 1024 : 1 MUX, how many 4 : 1 MUXs are required 341.

$$N = \frac{1024}{4} + \frac{256}{4} + \frac{64}{4} + \frac{16}{4} + 1$$

$$= 256 + 64 + 16 + 4 + 1$$

$$= 341$$

Question MCQ

★★☆

Which of the following is true:

A MUX is non-universal ckt. ✗

B ✓ MUX can be used to implement half adder as well as full adder.

C ✗ 2:1 MUX has 1–i/p line, 1–o/p line and 2-select lines.

D If in a MUX input lines are 24 then minimum 4-select lines are required for proper functioning.

$M \geq \log_2 N$ $\quad M \geq 4.8$

$M \geq \log_2 24$ $\quad M \geq 5$

Question MCQ

★★☆

Which of the following is true ?

A Decoder and MUX has same internal circuitry ✗

B MUX & De MUX has same internal circuitry ✗

C De MUX and decoder has same internal circuitry ✓

D None of these

Question **MCQ**

A combinational ckt using decoder and OR gate is as given below :

o/p y_1 & y_2 combinedly represents :



A Half adder

B Half subtractor with subtraction (B–A)

C Half subtractor with subtraction (A–B)

D Comparator o/p with (A > B)

Question NAT

★☆☆

To implement 1:512 De MUX, we require N, 1:8 De MUX, then the value of N is 73.

$$N = \frac{512}{8} + \frac{64}{8} + \frac{8}{8}$$

$$N = 64 + 8 + 1$$

$$\boxed{N = 73}$$

Question | MSQ | (B, C)

***

$1:128 = (1:2^7) \xrightarrow{1:2} (k^7 - 1) = 127$

To implement 1:128 De MUX, we require

A 127 – 1:4 DeMUX ✗

B ✓ 127 – 1:2 DeMUX

C ✓ 18 – 1:8 DeMUX & 1 – 1:2 DeMUX

D ✗ 40 – 1:8 DeMUX & 1 – 1:2 DeMUX

$1:128 \xrightarrow{1:4} \underbrace{32 + 8 + 2}$

$42\,(1:4)$

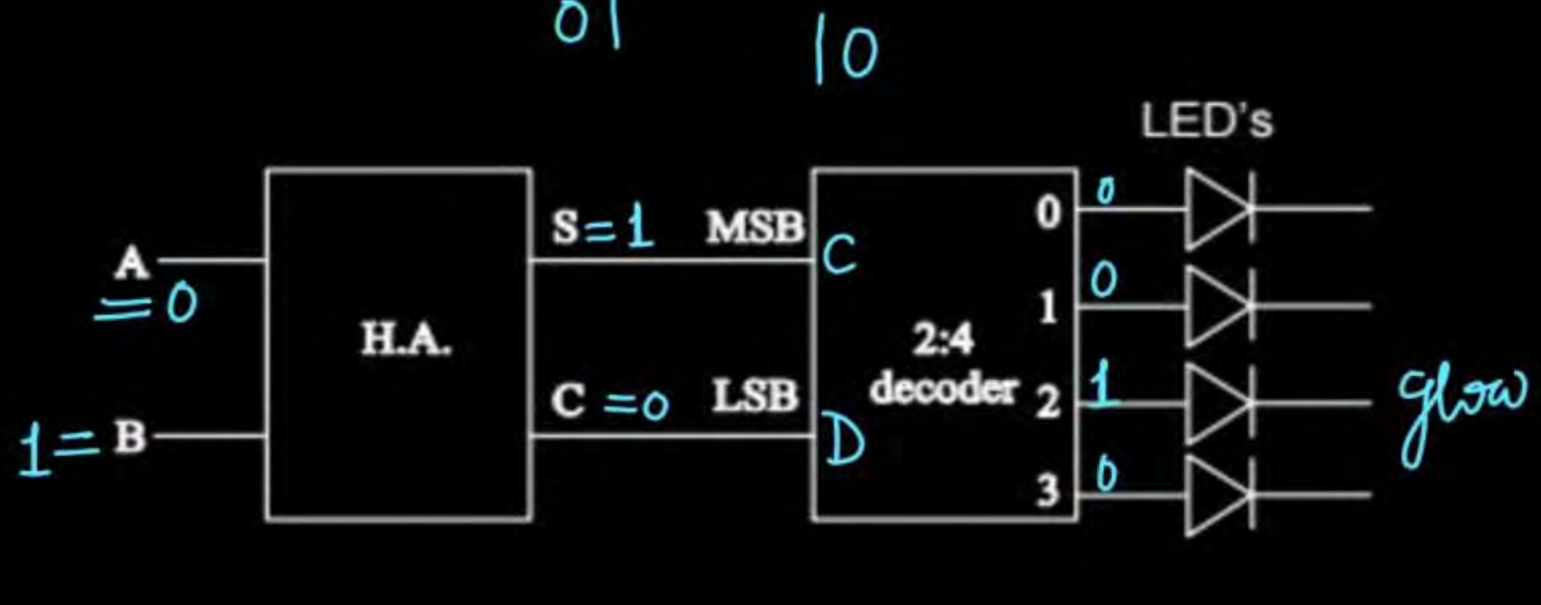
$+1\,(1:2)$

$1:128 \xrightarrow{1:8} \underbrace{16 + 2}$

$18\,(1:8) + 1\,(1:2)$

Question **NAT**

A combinational ckt is as given below
When i/p is A = 0 & B = 1, then LED no. that glows will be 2.



Question MCQ

* * *

For a decoder having 4 i/p lines, no. of o/p lines are

A 1

B 2

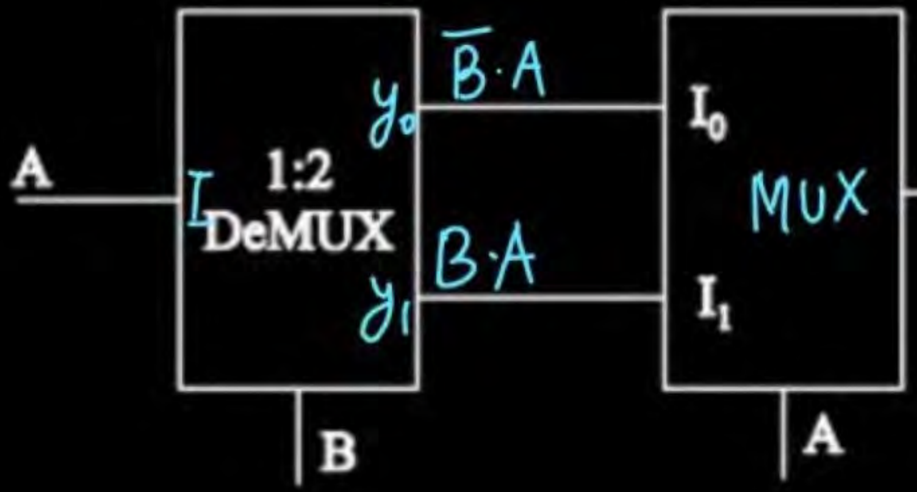
C 16

D 4

$n : 2^n$

Question MCQ

A digital ckt is as given below:



$y = \overline{A}\cdot A\overline{B} + A\cdot AB$

$= 0 + AB = AB$

o/p y is

A A

B B

C A.B

D (A + B)

Question MCQ

* * *

Which of the following is 'not true'?

$M \geq \log_2 N$

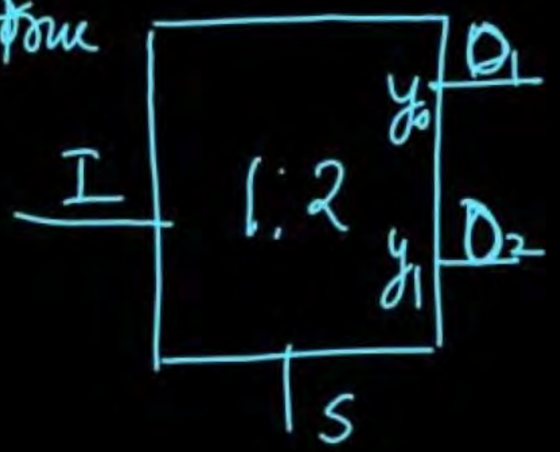
$M \geq \log_2 52 \rightarrow M \geq 5.\overline{W}$

A DeMUX is not a universal ckt → true

B For 52 output lines, no. of select lines must be atleast 6 for DeMUX. — true

C DeMUX is parallel to serial converter. not true

D DeMUX is serial to parallel converter true

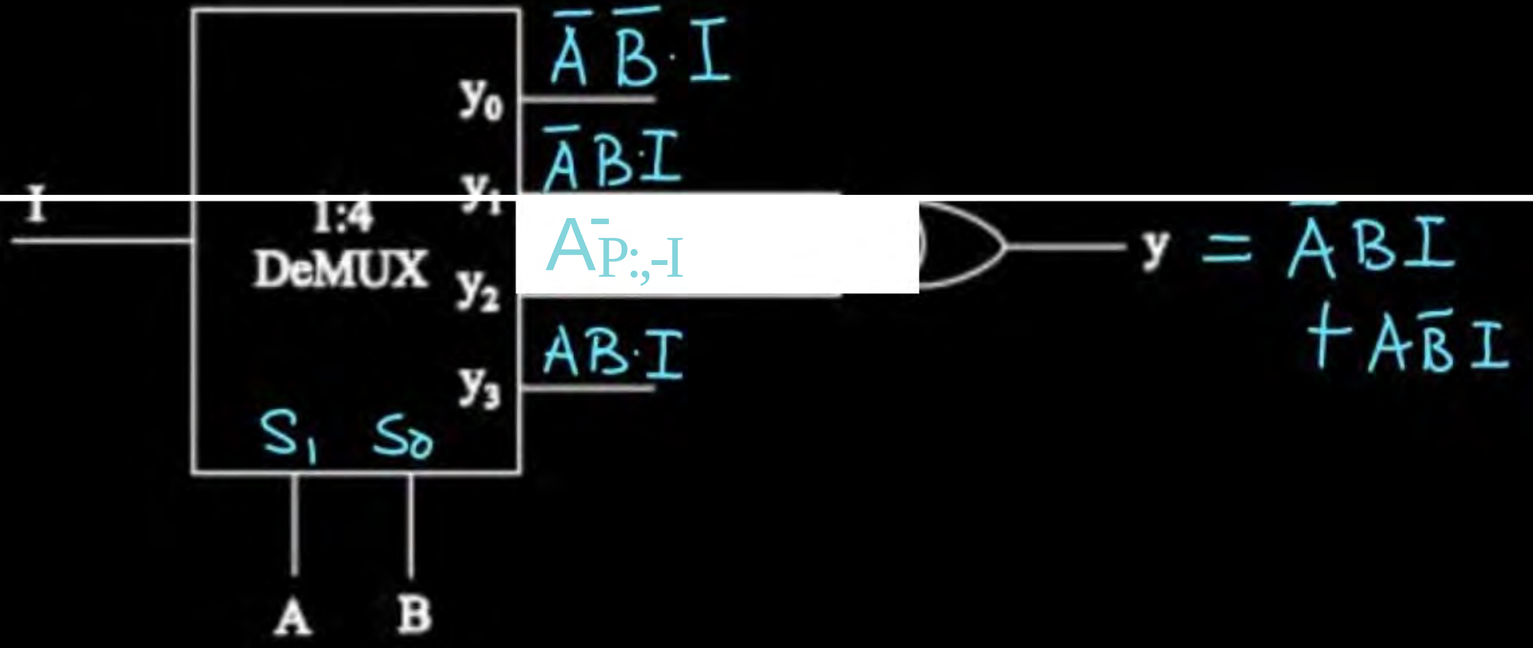


Question MCQ

* * *

A ckt is as shown below:

o/p y is



A $A \oplus B$ if $I = 1$

B $A \odot B$ if $I = 0$

C $A.B$ if $I = 1$

D $A + B$ if $I = 0$

Question NAT

★★★

For implementing full adder and full subtractor we require a decoder of the order n:M, where n is no. of i/p lines & M is no. of o/p lines, then value of M is 8.



F.A $\begin{cases} S = f(x,y,z) = \Sigma(1,2,4,7) \\ C = f(x,y,z) = \Sigma(3,5,6,7) \end{cases}$

F.S $\begin{cases} D = f(x,y,z) = \Sigma(1,2,4,7) \\ B = f(x,y,z) = \end{cases}$

Question **MSQ** (C)

no. of i/p lines = 4

4-i/p lines XNOR →

It is given that:

$x_1 \odot x_2 \odot x_3 \odot x_4 = 0$ ⟶ at i/p side odd no. of 1's are there ⟶ one 1's or three 1's

Then which of the following is/are true?

one i/p line is at '1' & other three are at '0'

1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

1010

A $x_1 x_2 + x_3 x_4 = 1$ ✗

B ✗ $x_1 \oplus x_2 + x_2 \oplus x_3 + x_3 \oplus x_4 = 1$

C ✓ Sum of x_1, x_2, x_3 & x_4 will be odd.

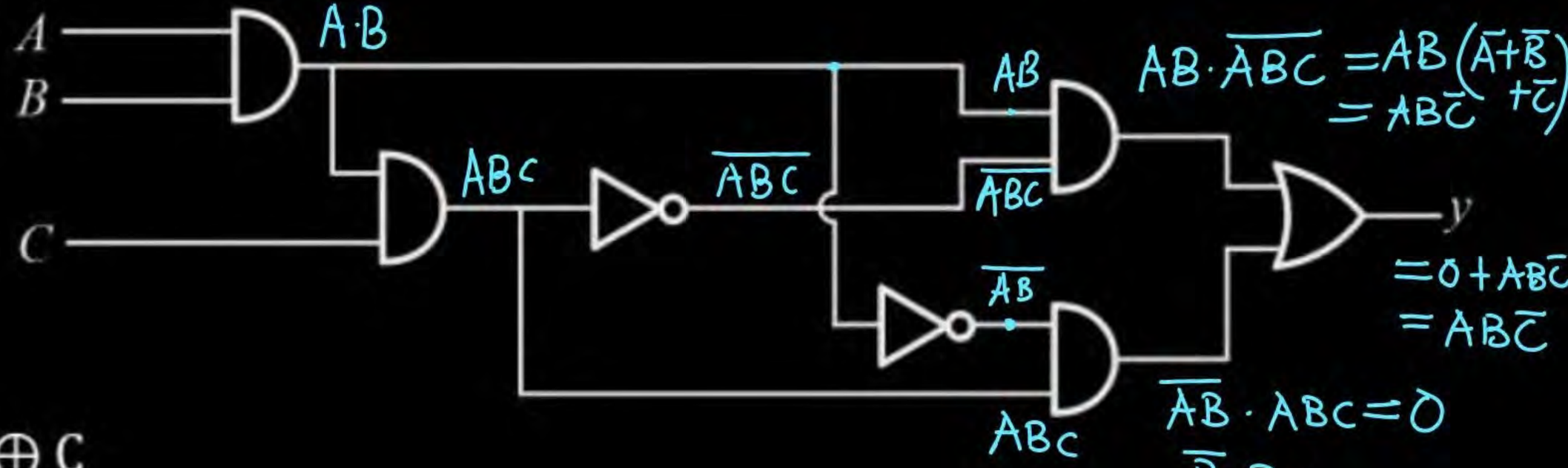
D ✗ $\overline{x_1 x_2} + x_2 x_3 + \overline{x_3 x_4} = 1$

Question

MCQ

A digital circuit is as given below:

Output y is



A $AB\bar{C}$

B $A \oplus B \oplus C$

C $A \odot B \oplus C$

D $\overline{A+B+C}$

Question

MCQ

$P = \overline{Carry} = \overline{xy + yz + zx}$

$Carry = \Sigma(3,5,6,7)$

A circuit is designed as shown:

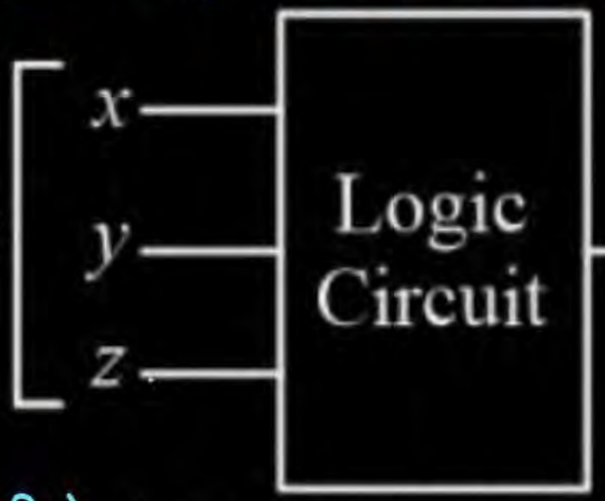
$= (\bar{x}+\bar{y})(\bar{y}+\bar{z})(\bar{z}+\bar{x})$

$P = \Sigma(0,1,2,4)$

Output P is 1 when majority of input lines are '0'.

Output P is

$= \bar{x}\bar{y} + \bar{y}\bar{z} + \bar{z}\bar{x}$



		Carry	P
	0 0 0	0	1
	0 0 1	0	1
	0 1 0	0	1
	0 1 1	1	0
	1 0 0	0	1

$= \overline{C}$

1 0 1	1	0
1 1 0	1	0
1 1 1	1	0

$(A+B)(B+C)(C+A)$

$AB + BC + CA = (A+B)(B+C)(C+A)$

A $xy + yz + zx$ ✗

B $\bar{x}\bar{y} + \bar{y}\bar{z} + \bar{z}\bar{x}$

C $\overline{x + y + z}$ ✗

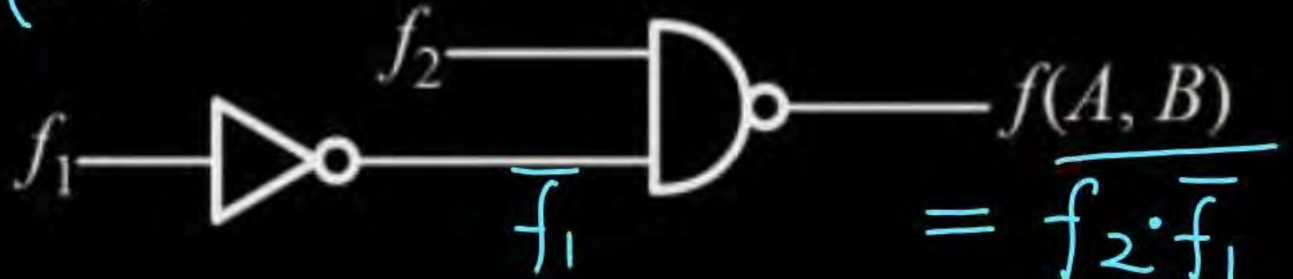
D $\overline{xy + yz}$ ✗

Question

MCQ

Two logical functions:
$f_1(A, B) = \Sigma(1, 2)$
$f_2(A, B) = \Sigma(0, 1, 3)$
then output f(A, B) will be

$\overline{f_2}(A,B) = \Sigma(2)$



$\overline{f(A,B)} = \overline{f_2 \cdot \overline{f_1}}$
$= \overline{f_2} + f_1$
$= \Sigma(1,2)$

A $\Sigma(1, 2)$

B $\Sigma(0, 1, 3)$

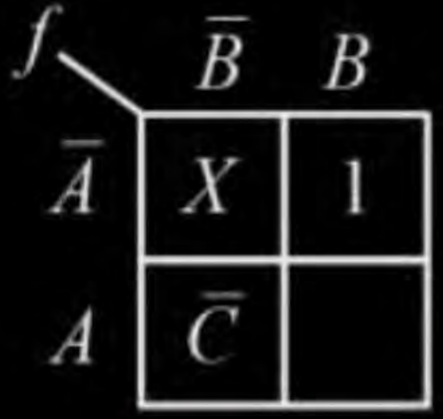
C $\Sigma(0, 1, 2, 3)$

D $\Sigma(1)$

Question

MCQ

A K-Map is as given below:
Its simplified solution will be



A $(\bar{A}+\bar{B})\bar{C}$

B $(\bar{A}+\bar{B})(\bar{A}+\bar{C})$

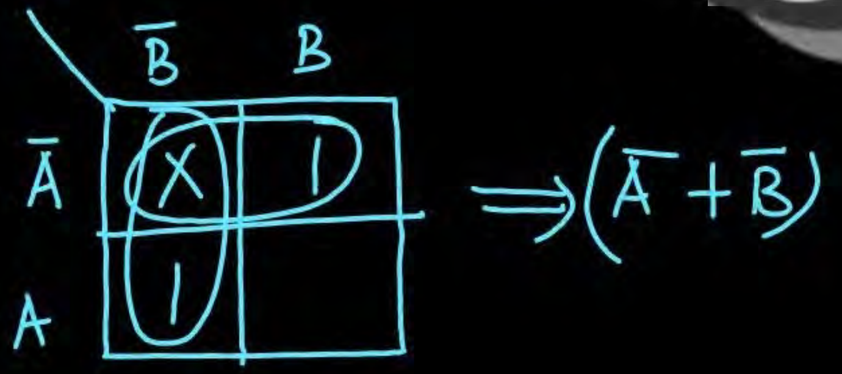
C $\overline{AB}+\overline{BC}$

D None of these

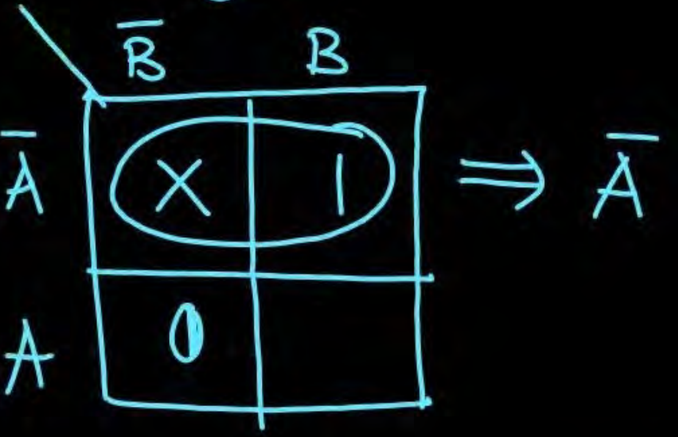
$= (\bar{A}+\bar{B})\bar{C} + \bar{A}\cdot C$

$= \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{A}C = \bar{A}(\bar{C}+C) + \bar{B}\bar{C} = \bar{A} + (\bar{B}\cdot\bar{C})$

$= (\bar{A}+\bar{B})\cdot(\bar{A}+\bar{C})$



C = 0

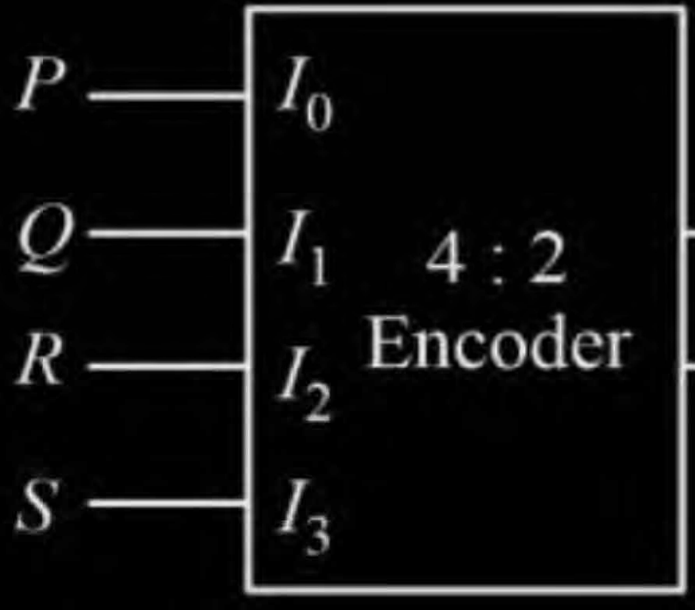


C = 1

Question MCQ

★☆☆

A 4 : 2 Encoder is as given below:
Output y_1 will be



I_0	I_1	I_2	I_3	y_1	y_0
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

$y_1 = I_2 + I_3 = (R+S)$

$y_0 = I_1 + I_3 = (Q+S)$

A (P + Q)

B (Q + R)

C (R + S)

D (P + S)

Question

MCQ

★★☆

Which of the following is not true?

A $A + \bar{B}C = (A + \bar{B})(\bar{B} + C)$ ⟹ not true

$A + (\bar{B} \cdot C) = (A + \bar{B}) \cdot (A + C)$

B $\bar{A} + \bar{B}C = \overline{AB} \cdot (\bar{A} + C)$ ⟶ true

$\bar{A} + (\bar{B} \cdot C) = (\bar{A} + \bar{B}) \cdot (\bar{A} + C) = \overline{A \cdot B}\ (\bar{A} + C)$

C $AB + \bar{A}C + \bar{B}C = AB + \bar{B}C$ true

$f\ AB + BC = (A + B)(B + C)$ not true

D $AB + BC + AC = (A + B)(B + C)(C + A)$

true

$f^D = (A + B) \cdot (B + C)$

Question NAT

For *n*-variable, the number of self dual functions are found to be 256, then the value of n is 4.

no. of boolean functions $= 2^{2^n} = 2^{2^4} = 2^{16} = 256 \times 256$

no. of self dual functions $= 2^{2^{n-1}} = 256 = 2^8$

$2^{n-1} = 8 = 2^3$

$n - 1 = 3$

$n = 4$

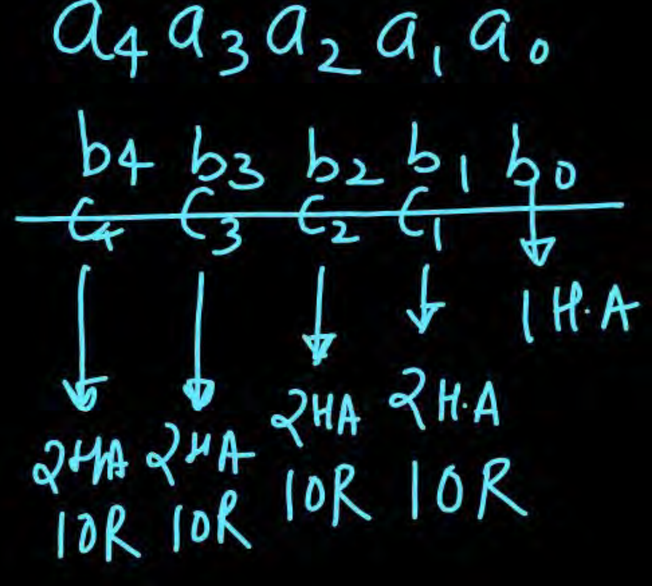
Question **NAT**



To add two 5 bit numbers, we required m H.A.s and n-OR gates then minimum value of (m +n) 13.

N → bit number ⇒

9 HA
4 OR



Question

MCQ

Two stage SOP circuit can directly be implemented

A Using NAND gates at first stage & AND gate at second stage.

B Using NAND gates only at both first & second stage.

C Using NOR gates only at both first & second stage.

D None of these

Thank you

GW

Soldiers !

